

Gourav Gulia

ML Engineer • MLOps • Generative AI

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Skills

Languages Python
Machine Learning Scikit-Learn, TensorFlow, XGBoost, Model Evaluation, Feature Engineering
Generative AI LangGraph, RAG, Prompt Engineering, Langchain
Computer Vision OpenCV, FaceNet512, RetinaFace, Image Quality Assessment (PSNR, SSIM, LBP)
MLOps Apache Airflow, MLflow, DVC, Docker, FastAPI, CI/CD
Software Eng. REST APIs, Model Serving, Dependency Injection, Factory Pattern, Python Packaging, Modular Architecture
Data & Libraries pandas, Pandas, NumPy, Pydantic
Databases SQL, MongoDB, VectorDB (Milvus, FAISS, Weaviate)
Tools Git, Linux, Streamlit

Experience

EY (Ernst & Young)

Jun 2025 - Jun 2026

Sr. Analyst

Gurugram

- Identified malpractice patterns and compliance gaps across large-scale government examination datasets using **pandas**, statistical analysis, and forensic validation techniques on multimodal data including facial images, demographic records, and system logs.
- Developed and deployed production-grade **face verification**, **image quality analysis**, and **morphing detection** pipelines for government clients including **SSC**, **HSSC**, and **NHA**, leveraging **FaceNet512**, **RetinaFace**, **PSNR**, **SSIM**, and **LBP** to improve verification accuracy across challenging image conditions.
- Designed a **weighted ensemble inference framework** optimized through **A/B Testing**, **LCB/UCB strategies**, and **ROC-based threshold tuning**, reducing false positives by nearly **11%** while improving verification consistency.
- Built an in-house **Text Similarity & Impersonation Detection System** using **Jaro-Winkler similarity**, **phonetic matching**, and **TF-IDF vectorization**, improving duplicate identity detection efficiency by approximately **25%** during forensic investigations.
- Re-architected legacy ML solutions into a modular, reusable, and API-driven framework using **FastAPI**, while implementing workflow orchestration with **Apache Airflow** and experiment tracking through **MLflow**, significantly improving deployment scalability and reproducibility.
- Optimized production inference pipelines through **multiprocessing**, **hyperparameter tuning**, and memory-efficient processing techniques including **dtype downcasting**, reducing inference latency by approximately **35%** and memory utilization by **22%**.
- Collaborated with cross-functional teams and government stakeholders to deliver enterprise **forensic AI solutions** aligned with regulatory compliance, risk governance, and standardized investigation workflows across large-scale public sector projects.

Netmax

Oct 2024 - Feb 2025

Jr. Data Scientist

Chandigarh

- Built a modular data preprocessing pipeline leveraging **MLflow**, **DVC**, and **Pandas**, which reduced data drift and noise by 20%, boosting downstream model accuracy across production workflows.

Education

M.Sc. Data Science

Jul 2022 – May 2024

Chandigarh University

B.Sc. Applied Science

Jul 2019 – May 2022

Delhi University

Featured Projects

Aurelius - Autonomous Multi-Agent Research & LLMOps System | Links •

Jun 2026 - Aug 2026

- Architected a stateful **multi-agent research workflow** using LangGraph and LangChain with 5 specialized personas: Lead, Researcher, Synthesizer, Writer, and Reviewer, enabling autonomous query decomposition, cross-source evidence gathering, and self-correcting revision loops for citation-backed technical reports.
- Engineered an enterprise-grade **LLMOps observability and evaluation framework** with LangSmith distributed tracing, granular token and latency tracking for Groq LPU inference, and automated quality scorecards measuring citation density, factual consistency, and evidence grounding with a strict zero-link hallucination guarantee.
- Built a production-ready **Python 3.12 architecture** using Pydantic v2 validation, provider-abstracted LLM factories, resilient search and scraping fallback pipelines, and structured state management for reliable multi-agent execution.
- Developed a real-time **Streamlit reasoning dashboard** and production containerization workflow using multi-stage Docker and UV, backed by a **28-test automated test suite** covering core workflow, validation, and reliability components.

End-to-End AI-Powered Burnout Risk Prediction | Links • • • • •

Jun 2026 - Aug 2026

- Engineered an end-to-end **student burnout prediction system** covering data validation, feature engineering, model training, evaluation, and production inference.
- Built a configurable **ML experimentation framework** for systematic model comparison and hyperparameter optimization across multiple classification algorithms.
- Implemented a **reproducible, artifact-driven ML workflow** using DVC and MLflow for versioned datasets, experiment tracking, model management, and pipeline reproducibility.
- Productionized the model with **FastAPI, Streamlit, and Docker**, deploying a live prediction API and interactive UI with a publicly available container image.

CIP: Candidate Intelligence Platform

Jan 2026 - Mar 2026

- Developed a **Ensembled candidate risk prediction model** using structured examination metadata and forensic indicators, providing the foundation for downstream explainable AI workflows.
- Architected a modular, configuration-driven **AI risk intelligence platform** at EY GPS Assurance using reusable pipeline components, artifact-based communication, **Dependency Injection**, and YAML-driven configurations, enabling scalable and maintainable ML workflows.
- Built a production-grade **MLOps pipeline** leveraging **Apache Airflow, DVC, MLflow, Docker, and FastAPI** for automated orchestration, experiment tracking, data versioning, reproducible model training, and scalable inference.
- Developed an AI-powered **Investigation Copilot** using **LangGraph**, advanced **prompt engineering**, and a **Retrieval-Augmented Generation (RAG)** architecture to retrieve SOPs, historical investigation cases, and policy documents for explainable, evidence-backed decision support.
- Designed an interactive **Streamlit** investigation dashboard integrating real-time ML predictions with conversational AI, enabling investigators to analyze candidate risk, understand model reasoning, and receive context-aware investigation recommendations through a production-ready interface.

Inspector: In-House Forensic Analytics Library for EY

Jul 2025 – Aug 2025

- Designed an in-house modular **Python package (.whl)** and **Dockerized forensic analytics library** at EY for scalable image and text-based ML workflows, enabling reusable and production-ready forensic pipelines.
- Engineered a high-performance **face retrieval system** using **VectorDB** and **HNSW-based ANN indexing**, enabling near real-time similarity search across **1M+ embeddings** with optimized retrieval latency.
- Built a scalable **text forensic analytics engine** using **Jaro-Winkler similarity, TF-IDF vectorization, and nearest neighbor search** to detect impersonation and duplicate identity patterns across large-scale datasets.
- Accelerated ML pipeline performance using **multiprocessing, memory-efficient processing**, and advanced software engineering principles including **Dependency Injection** and **Factory Pattern**, improving maintainability and execution efficiency.

Research Publications

Liver Disease Prediction Using Ensemble Learning

- Developed a predictive system for **liver disease detection** using multiple **machine learning algorithms**, enabling early-stage diagnosis through data-driven insights.
- Performed **comparative analysis** across models to evaluate performance using metrics such as **accuracy, precision, and recall**, achieving up to **95% accuracy**.
- Applied comprehensive **data preprocessing, feature selection, and model validation** techniques to improve generalization and robustness on medical datasets.
- Demonstrated strong understanding of **healthcare analytics, statistical modeling, and ML evaluation frameworks**, contributing to a research-backed solution for disease prediction.